



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

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CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No : 5

TITLE :Develop aJavaapplication demonstrating the use of ArrayList, Vector, and StringBufferfor dynamicdatastorage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

```
import java.util.ArrayList;
import java.util.Vector;

public class Main {
    Run | Debug
    public static void main(String[] args) {

        // ArrayList
        ArrayList<String> students = new ArrayList<>();
        students.add(e: "Dharmit");
        students.add(e: "Ishaan");
        students.add(e: "Pranil");

        System.out.println(x: "ArrayList:");
        for (String s : students) {
            System.out.println(s);
        }

        // Vector
        Vector<Integer> marks = new Vector<>();
        marks.add(e: 85);
        marks.add(e: 90);
        marks.add(e: 78);

        System.out.println(x: "\nVector:");
        for (int m : marks) {
            System.out.println(m);
        }

        // StringBuffer
        StringBuffer message = new StringBuffer(str: "Java");
        message.append(str: " Programming");
        message.append(str: " Lab");

        System.out.println(x: "\nStringBuffer:");
        System.out.println(message);
    }
}
```

```

PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp5> java Main.java
ArrayList:
Dharmit
Ishaan
Pranil

Vector:
85
90
78

StringBuffer:
Java Programming Lab
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp5>

```

Exercise

Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

```

import java.util.ArrayList;

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        ArrayList<String> arr = new ArrayList<>();

        arr.add(e: "Study");
        arr.add(e: "Play a sport");
        arr.add(e: "Coding");
        arr.add(e: "Build a project");
        arr.add(e: "Nap");

        StringBuffer str = new StringBuffer();

        for (int i = 0; i < arr.size(); i++) {
            str.append((i + 1) + "." + arr.get(i) + "\n");
        }

        System.out.println(x: "----- TO DO List -----");
        System.out.println(str);
    }
}

```

```

PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp5\Exercise> java Ex1.java
----- TO DO List -----
1.Study
2.Play a sport
3.Coding
4.Build a project
5.Nap

```

Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
import java.util.ArrayList;

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        ArrayList<String> courses = new ArrayList<>();

        courses.add(e: "Java");
        courses.add(e: "Python");
        courses.add(e: "Data Structures");
        courses.add(e: "Operating Systems");

        courses.remove(o: "Python");

        StringBuffer sb = new StringBuffer();

        sb.append(str: "Registered Courses\n");
        sb.append(str: "-----\n");

        for (int i = 0; i < courses.size(); i++) {
            sb.append((i + 1) + ". " + courses.get(i) + "\n");
        }

        System.out.println(sb);
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp5\Exercise> java Ex2.java
Registered Courses
-----
1. Java
2. Data Structures
3. Operating Systems
```